

SDF architecture Unless otherwise stated, the learned SDFs consisted of MLPs with 8 layers and 128 hidden units. The models were trained for 400K steps and a learning rate of 10^{-5} for Kubric Meshes. For the Garden Vase scene, MLPs with 8 layers and 256 units were used, but training details were otherwise the same. For quantitative results in Figure 6, we run the all the model with 3 seeds and report the mean and the 95% confidence interval.

SDF training time and hardware For all meshes in Movi-C, we trained the MLPs for 400k iterations, which requires 18 hours of training time on average on a single V100 GPU. It is likely that this training time could be reduced, as we did not aim to minimize the computational cost of the SDFs in this work. Instead, we chose to train the SDF to maximize the quality for complex Movi-C meshes that have 8,982 nodes and 15,955 triangles on average.

Other approaches tested In NeRF literature a common regularisation to train SDFs is the Eikonal regularisation that enforces unit-norm of the SDF gradient. However, we found that Eikonal regularisation leads to over-smoothed sphere-like shapes and did not use it in our training. Furthermore, we find that our approach of directly fitting the distances already results in unit-norm-gradient and does not require an additional regularisation term. We also tested another common technique of periodic encodings as the MLP inputs, however, we found that it leads to checkerboard-like artifacts for points outside of the object and did not use this method.

D.2 Training SDFs from vision

For an experiment in section 4.4, we take a Garden Scene from [4], represented as a set of 2D images with a 360-degree view of the scene. We train a VolSDF model [39] to distill an SDF from this scene. VolSDF leverages a volumetric representation that explicitly encodes the volume density as a transformed SDF. This explicit formulation provides an inductive bias over the geometry of an object and a built-in preference for capturing smooth and clean shapes.

VolSDF already represents a learned SDF and, in principle, can be directly used in SDF-Sim. However, to simplify the pipeline and have a consistent SDF architecture for all objects, we train another learned SDF, as described in Section D. To do so, we extract a high quality mesh (with over 80k vertices) of the table and the vase from VolSDF using the Marching Cubes algorithm [16]. Then we train a learned SDF with the architecture described in section D and use it in the SDF-Sim.

E Additional results

E.1 Quantitative Results on small benchmark datasets.

Movi-B

Model	Translation RMSE (m)	Rotation Err (deg)
DPI	0.368 ± 0.057	26.928 ± 2.740
MGN-LargeRadius	0.460 ± 0.045	26.342 ± 1.397
MGN	0.538 ± 0.035	26.914 ± 0.783
FIGNet	0.14 ± 0.0087	15.0 ± 0.675
FIGNet*	0.13 ± 0.008	15.94 ± 0.8849
SDF-Sim (ours)	0.16 ± 0.0124	18.03 ± 0.9433

Model	# Collision Edges	# Graph Edges	Peak Memory (MiB)	Runtime per step (ms)
DPI	2250.688 ± 65	-	-	0.145 ± 0.009
MGN-LargeRadius	1797.985 ± 60	-	-	0.218 ± 0.033
MGN	34.367 ± 4	-	-	0.175 ± 0.018
FIGNet	1360.99 ± 196	24569.59 ± 938	94.45 ± 2.9698	0.23 ± 0.0044
FIGNet*	1409.87 ± 267	8613.05 ± 512	63.83 ± 3.7489	0.22 ± 0.0077
SDF-Sim (ours)	201.14 ± 10	6195.58 ± 193	38.8 ± 0.5711	0.14 ± 0.004

Table 3: Quantitative comparison of SDF-Sim and the baselines on a **Movi-B** dataset. This table presents the numbers for the Figure 6 in the main text. The error bars show the 95% confidence interval from running all models with 3 random seeds. DPI, MGN-LargeRadius and MGN results are reported by [1]. Results marked as '-' were not reported by the previous work [1]

Movi-C

Model	Translation RMSE (m)	Rotation Err (deg)
DPI-Reimplemented*		OOM
MGN-LargeRadius*		OOM
MGN*		OOM
FIGNet		OOM
FIGNet*	0.18 ± 0.0097	19.68 ± 0.6483
SDF-Sim (ours)	0.24 ± 0.0151	23.15 ± 0.668

Model	# Collision Edges	# Graph Edges	Peak Memory (MiB)	Runtime per step (ms)
DPI-Reimplemented*			OOM	
MGN-LargeRadius*			OOM	
MGN*			OOM	
FIGNet			OOM	
FIGNet*	2220.96 ± 469	11502.18 ± 944	75.87 ± 9.587	0.30 ± 0.0163
SDF-Sim (ours)	300.46 ± 34	7661.17 ± 266	43.35 ± 0.7466	0.17 ± 0.0038

Table 4: Quantitative comparison of SDF-Sim and the baselines on a **Movi-C** dataset. This table presents the numbers for the Figure 6 in the main text. The error bars show the 95% confidence interval from running all models with 3 random seeds. DPI, MGN-LargeRadius and MGN and FIGNet baselines run OOM on the Movi-C simulations and are not reported.

E.2 Additional metrics on SDF quality

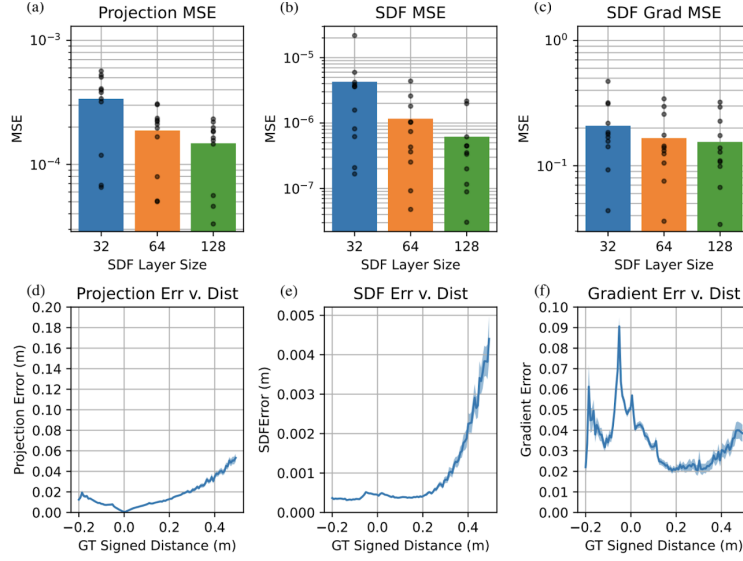


Figure S1: **Additional metrics on SDF quality**

(a-c) Metrics computed across learned SDF models of different sizes for different objects in Movi-B. Each black marker denotes a different Movi-B shape. For each metric, loss decreases as we increase the size of the model.

(a) MSE between the projected surface points predicted from the learned SDFs and the true closest point on the surface.

(b) MSE between the signed distance outputted by the learned SDF and the true signed distance.

(c) MSE between the gradient of the learned SDF and the gradient of the ground truth SDF function.

(d-f) **SDF evaluation metrics as a function of distance from the object surface.** SDF error, projection error and the gradient errors remain constant up to distance 0.2 from the object surface and start to increase afterwards. Generally all errors remain low in comparison to the object size of 0.5 meter.

(d) Error as distance between the learned SDF projected surface point and the true closest surface point, plotted as a function of signed distance from the surface.

(e) Error as the absolute difference in magnitude between the learned SDF prediction and ground truth SDF, plotted as a function of signed distance from the surface.

(f) Error as the cosine distance between the gradient of learned SDF and the ground truth SDF gradient.